

# XIAOFENG TAN

Shenzhen, Guangdong, China

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## RESEARCH INTERESTS

RLHF, RLVR, AIGC, 3D Human Modeling, Preference Optimization, Diffusion Models, Multimodal Learning.

## EDUCATION

**Southeast University (SEU)**, Nanjing, China

Sept 2024 – Present

CSRankings (China, 2016–2026): AI #16 · CV #25 · ARWU 2026 #89

M.Sc. in Computer Science [PALM Lab](#), supervised by Prof. [Hongsong Wang](#)

GPA: **88.6**/100

First-Class Academic Scholarship (top 10%) & Tencent-SEU Scholarship (top 2.5%); Summer Camp Rank 2

**Shenzhen University (SZU)**, Shenzhen, China

Sept 2020 – Jun 2024

CSRankings (China, 2016–2026): AI #21 · CV #20 · ARWU 2026 151–200

B.E. in Computer Science & B.Sc. in Mathematics (*Dual Degree*)

GPA: **88.9**/100 · **4.07**/4.50

Outstanding Graduate Representative (1/300+) and graduation speaker; featured in [SZU's 2024 Admissions Guide](#)

## SELECTED PUBLICATIONS

\* indicates equal contribution or first author (student.)

### Published

**When Policy Entropy Constraint Fails: Preserving Diversity in Flow-based RLHF via Perceptual Entropy**

Xiaofeng Tan, Jun Liu, Bin-Bin Gao, Yuanting Fan, Xi Jiang, Chengjie Wang, Hongsong Wang, Feng Zheng

[NeurIPS'26](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A\*)

[Spotlight](#) (292/30,709 submissions, 0.95%)

[\[arXiv\]](#) [\[Project\]](#)

**EasyTune: Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation**

Xiaofeng Tan\*, Wanjiang Weng\*, Haodong Lei, Hongsong Wang

[ICLR'26](#) · *International Conference on Learning Representations* (CCF-A, CORE-A\*)

[\[PDF\]](#) [\[Project\]](#) [\[Openreview\]](#)

**ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment**

Wanjiang Weng\*, Xiaofeng Tan\*, Junbo Wang, Guo-Sen Xie, Pan Zhou, Hongsong Wang

[AAAI'26](#) · *AAAI Conference on Artificial Intelligence* (CCF-A, CORE-A\*)

[\[PDF\]](#) [\[Project\]](#) [\[Code\]](#)

**FG-Diff: Frequency-Guided Diffusion Model with Perturbation Training for Skeleton-Based VAD**

Xiaofeng Tan, Hongsong Wang, Xin Geng, Liang Wang

[TIP'26](#) · *IEEE Transactions on Image Processing* (CCF-A, SCI-Q1)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

**SoPo: Text-to-Motion Generation Using Semi-Online Preference Optimization**

Xiaofeng Tan, Hongsong Wang, Xin Geng, Pan Zhou

[NeurIPS'25](#) · *Advances in Neural Information Processing Systems* (CCF-A, CORE-A\*)

[\[Project\]](#) [\[PDF\]](#) [\[Code\]](#)

**Fuzzy Granule Density-Based Outlier Detection with Multi-Scale Granular Balls**

Can Gao (Supervisor), Xiaofeng Tan\*, Jie Zhou, Weiping Ding, Witold Pedrycz

[TKDE'25](#) · *IEEE Transactions on Knowledge and Data Engineering* (CCF-A, SCI-Q1)

[\[PDF\]](#) [\[Code\]](#)

### Preprints

**MotionRFT: Unified Reinforcement Fine-Tuning for Text-to-Motion Generation**

Xiaofeng Tan, Wanjiang Weng, Hongsong Wang, Fang Zhao, Xin Geng, Liang Wang

*Under Review (Submitted to TPAMI) (Under Second-Round Review)*

[\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)

**Dual-Grained Policy Optimization for Hallucination Reduction in Flow-based RL**

Xiaofeng Tan, Jun Liu, Yuanting Fan, Bin-Bin Gao, Xi Jiang, Xiaochen Chen, Jinlong Peng, Chengjie Wang, Hongsong

Wang, Feng Zheng

RESEARCH EXPERIENCE

|                                                                                                                                                                                            |                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Tencent Inc.</b> , Shenzhen, China<br>Research Intern @ <a href="#">Hunyuan</a><br>Research Direction: RLHF, RLVR, AIGC                                                                 | Mar 2026 – Present<br>Mentor: Jiaxin Lin                                                              |
| <b>University of California, Merced</b> , Merced, CA (Remote)<br>Remote Visiting Student @ <a href="#">Vision and Learning Lab (VLL)</a><br>Research Direction: Computer Vision            | Apr 2026 – Present<br>Mentor: Ming-Hsuan Yang                                                         |
| <b>Tencent Inc.</b> , Shenzhen, China<br>Research Intern @ <a href="#">YouTu Lab</a><br>Research Direction: RLHF, RLVR, AIGC                                                               | Jul 2025 – Mar 2026<br>Mentor: Jun Liu<br>Works: Consistent-RFT, PEC-RFT (NeurIPS'26 Spotlight)       |
| <b>Singapore Management University</b> , Singapore (Remote)<br>Remote Visiting Student @ <a href="#">Language and Vision Lab (LV Lab)</a><br>Research Direction: RLHF, Motion Generation   | Jul 2024 – Mar 2025<br>Mentor: Pan Zhou<br>Works: SoPo (NeurIPS'25), ReAlign (AAAI'26)                |
| <b>Southeast University</b> , Nanjing, China<br>Master Student @ <a href="#">PAttern Learning and Mining (PALM) Lab</a><br>Research Direction: RLHF, Motion Generation, Action Recognition | Apr 2024 – Present<br>Mentor: Hongsong Wang<br>Works: FG-Diff (TIP'26), EasyTune (ICLR'26), MotionRFT |
| <b>Shenzhen University</b> , Shenzhen, China<br>Undergraduate Research Assistant @ <a href="#">Computer Vision Institute</a><br>Research Direction: Outlier Detection, Anomaly Detection   | Oct 2022 – Oct 2023<br>Mentor: Can Gao<br>Works: MGBOD (TKDE), TWCOD (IJAR)                           |

SERVICE

**Conference Reviewer:** ICML 2026 (*Gold Reviewer*), NeurIPS 2026, AAAI 2027, ICLR 2027  
**Journal Reviewer:** Transactions on Machine Learning Research (TMLR), International Journal of Data Science and Analytics (IJDSA)

HONORS & AWARDS

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|-----------------------------------------------------|-----------------------------------------------------------|------------|
| <b>ICML Gold Reviewer</b>                           | International Conference on Machine Learning              | 2026       |
| <b>Tencent-SEU Scholarship (top 2.5%)</b>           | Southeast University (¥9,000)                             | 2025       |
| <b>First-Class Academic Scholarship (top 10%)</b>   | Southeast University (¥12,000)                            | 2025       |
| <b>Outstanding Graduate Representative</b>          | Shenzhen University (Graduation Speech)                   | 2024       |
| <b>Featured in Admissions Guide</b>                 | Shenzhen University (as Outstanding Student)              | 2024       |
| <b>Honor Bachelor Degree (top 3%)</b>               | Shenzhen University                                       | 2024       |
| <b>Outstanding Graduate (top 5%)</b>                | Shenzhen University                                       | 2024       |
| <b>LiYuan Star (SZU Special Award, top 0.73%)</b>   | Shenzhen University (¥20,000)                             | 2022, 2023 |
| <b>Scholarship of Outstanding Innovative Talent</b> | Guangdong Science College Entrance Exam (top 2%, ¥20,000) | 2020       |